

TRAN THUAN HOAN

UNDERGRADUATE RESEARCHER · COMPUTATIONAL ECONOMICS · QUANTITATIVE METHODS

Hanoi, Vietnam · tthoan309@gmail.com · [LinkedIn](#) · [GitHub](#) · [ORCID 0009-0007-6229-8585](#)

RESEARCH PROFILE

International Economics undergraduate combining econometrics, machine learning, data engineering, and research software to study complex economic systems. Builds reproducible evidence pipelines—from data acquisition and normalization to modeling, interpretation, and evaluation—with current interests in energy transition, green growth, financial risk, trade networks, and AI for economics.

EDUCATION

Foreign Trade University (FTU)

Expected 2027

Bachelor of International Economics · High Quality Program · GPA: 3.72 / 4.00

Selected coursework: Advanced Mathematics (10.0); Probability & Statistics (9.3); Econometrics I (9.5); Econometrics II (10.0)

English: IELTS 7.0

CURRENT WORKING PAPER

Dynamic Relationship between Energy Transition and Green Growth

In development

The Moderating Role of Public Attention

- Developing a dynamic empirical framework to test how energy transition shapes green growth and when public attention changes the strength of that relationship.
- Current stage: literature review and data construction; model development and writing follow measurement validation.

SELECTED RESEARCH SYSTEMS

Financial Distress Prediction Pipeline [Repository](#) ↗

2026 · In development

End-to-end financial research infrastructure and explainable early-warning modeling

- Automates CafeF crawling and financial-statement collection; extracts irregular disclosures through PDF parsing and OCR with traceable source metadata.
- Normalizes accounting labels, units, periods, and identities into a versioned firm-level dataset; engineers liquidity, leverage, profitability, cash-flow, and trajectory features.
- Evaluates distress classifiers with temporal validation, rare-event metrics, calibration, leakage checks, and SHAP explanations; preserves extraction and modeling decisions as auditable research artifacts.

ESG-RAG Index [Repository](#) ↗

2026 · Prototype

Evidence-grounded retrieval infrastructure for corporate ESG disclosures

- Builds a layout-aware ingestion, parsing, metadata, chunking, and retrieval pipeline for sustainability reports while preserving issuer, page, period, and extraction-version provenance.
- Prioritizes citation auditability and retrieval evaluation over unsourced generation, enabling reproducible comparison across firms and reporting periods.

Macroeconomic Crisis Early Warning System

2025 · Completed

Explainable machine-learning system for policy monitoring

- Frames crisis detection as a rare-event classification problem and compares linear and nonlinear models using discrimination, calibration, and class-sensitive metrics.
- Uses SHAP to inspect system-level vulnerability drivers and country-level alerts, connecting predictive performance with policy-relevant interpretation.

ADDITIONAL RESEARCH EVIDENCE

Economic Sanctions & Trade Network Reconfiguration <i>International trade, panel econometrics, and network science</i> Combines product-level bilateral trade flows, sanctions events, gravity-style estimation, and network measures to separate lost trade from rerouting and partner substitution.	2026 · Research design
Inventory Optimization under Demand Uncertainty Repository ↗ <i>Stochastic decision science and simulation-based policy evaluation</i> Compares replenishment policies across Monte Carlo demand regimes and stress-tests service, holding-cost, shortage-risk, volatility, and lead-time assumptions.	2026 · Model developed
Vietnam–EU Monthly CN8 Trade Dataset <i>Reproducible product-level trade research infrastructure</i> Automates acquisition, code concordance, schema validation, unit harmonization, duplicate checks, and versioned release generation for monthly CN8 trade flows.	2026 · Construction

RESEARCH EXPERIENCE & LEADERSHIP

Vice President · International Economics Union (IEU) Coordinate research-oriented academic activities and interdisciplinary collaboration for the International Economics student community.	Current
Research & Development Department · Young Researchers Club, FTU Develop research concepts, review analytical approaches, and contribute to workshops, seminars, and student-led empirical research activities.	Current
Research Department · Global Economics Club, FTU Support evidence synthesis, quantitative analysis, and collaborative discussion of policy-relevant economic questions.	Current
Data Analysis Division · G'Contest Organizing Committee Supported a university-level data analysis competition with 500+ participants; coordinated analytical and operational tasks across the organizing team.	2025

METHODS & RESEARCH TOOLS

Econometrics: Panel data; fixed/random effects; PPML; gravity models; specification diagnostics; quantitative and policy analysis	Research infrastructure: Web crawling; OCR; PDF parsing; data normalization; validation pipelines; versioned dataset construction
Machine learning: Logistic regression; random forest; XGBoost; regression; SHAP; calibration; temporal and comparative evaluation	Optimization & networks: ALNS; Monte Carlo simulation; network centrality; community detection; graph embeddings
Programming: Python; SQL; Stata; Pandas; NumPy; Scikit-learn	Decision support: Power BI; DAX; Matplotlib; KPI design; interactive dashboards; data storytelling